

Computational Wind Engineering

Finite differences

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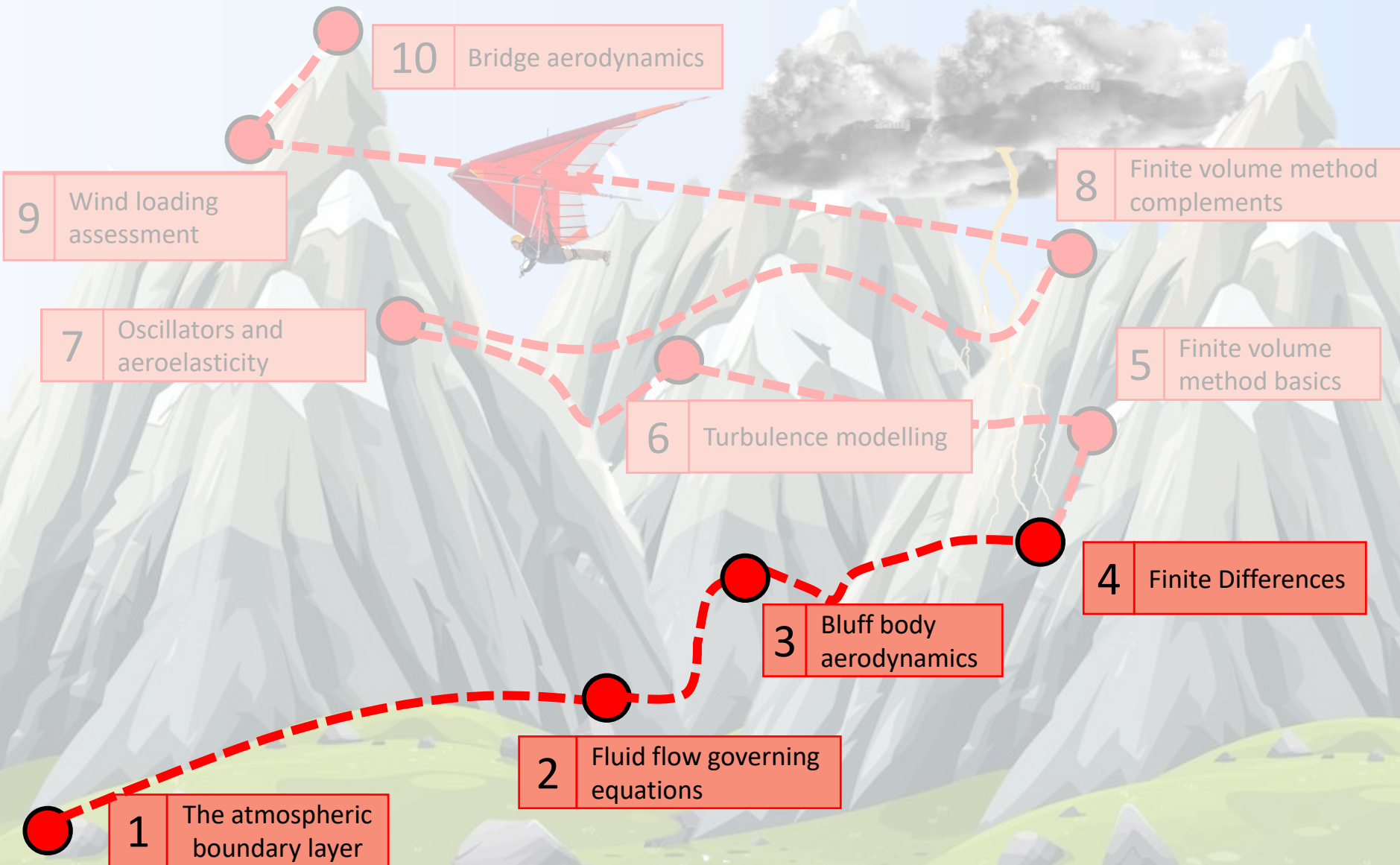
- The advection-diffusion equation
- Approximation schemes
- Analysis of the truncation error
- Numerical viscosity
- Origins of the Finite Volume method

Map of the course

Wind Engineering

Fluid Dynamics

Numerical methods



We now start to consider the necessity to **obtain solutions of the governing equations**, i.e. Navier-Stokes. To address the problem, we consider the simplest case, so we simplify

$$\begin{cases} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \\ \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}u + \frac{\partial u}{\partial y}v + \frac{\partial u}{\partial z}w + \frac{1}{\rho}\frac{\partial p}{\partial x} - \frac{\mu}{\rho}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) = \frac{f_x}{\rho} \\ \frac{\partial v}{\partial t} + \frac{\partial v}{\partial x}u + \frac{\partial v}{\partial y}v + \frac{\partial v}{\partial z}w + \frac{1}{\rho}\frac{\partial p}{\partial y} - \frac{\mu}{\rho}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right) = \frac{f_y}{\rho} \\ \frac{\partial w}{\partial t} + \frac{\partial w}{\partial x}u + \frac{\partial w}{\partial y}v + \frac{\partial w}{\partial z}w + \frac{1}{\rho}\frac{\partial p}{\partial z} - \frac{\mu}{\rho}\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right) = \frac{f_z}{\rho} \end{cases}$$

We then consider a mono-dimensional case

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}u - \nu \frac{\partial^2 u}{\partial x^2} = 0.$$

We further proceed with the usual decomposition

$$u(x, t) = U + u'(x, t).$$

Which substituted into the governing equation

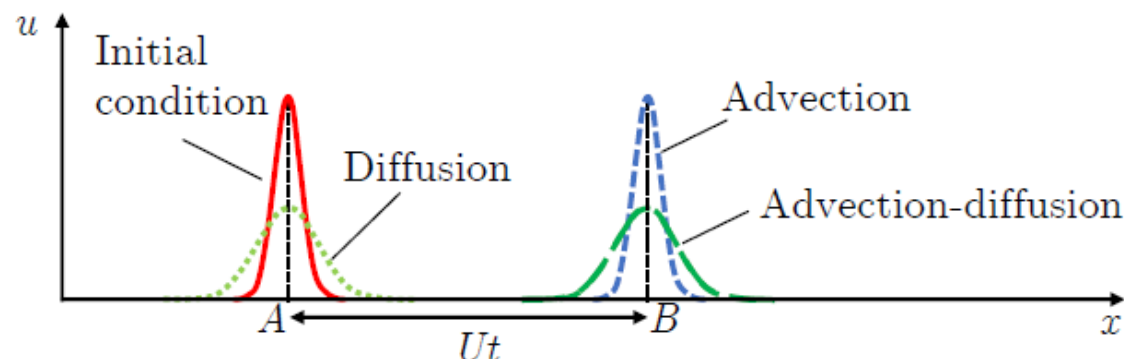
$$\frac{\partial(U + u'(x, t))}{\partial t} + \frac{\partial(U + u'(x, t))}{\partial x}(U + u'(x, t)) - \nu \frac{\partial^2(U + u'(x, t))}{\partial x^2} = 0,$$

and assuming U to be uniform (same value everywhere in space)

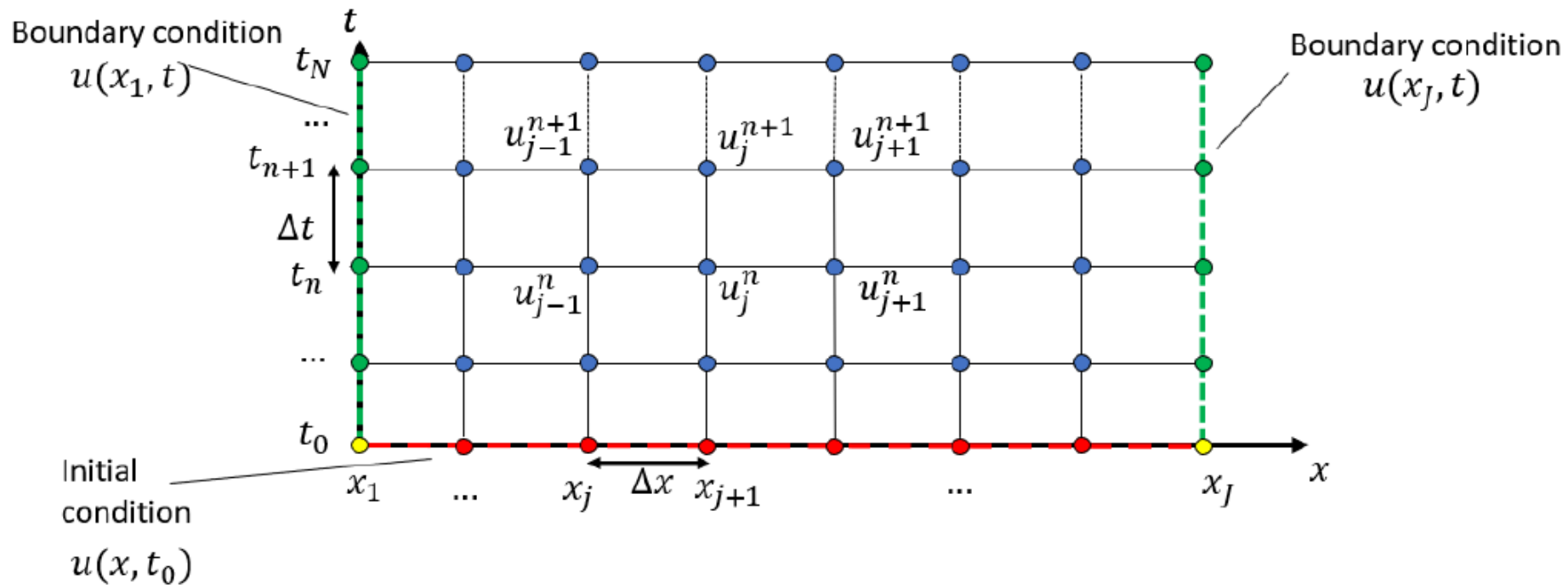
$$\frac{\partial u'}{\partial t} + \frac{\partial u'}{\partial x}(U + u') - \nu \frac{\partial^2 u'}{\partial x^2} = 0.$$

which assuming $U \gg u'$ and suppressing the apex on fluctuating quantities leads to

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} - \nu \frac{\partial^2 u}{\partial x^2} = 0,$$



We start by considering a grid of points in space-time, useful to calculate our solution.



Our idea is to write an equation at each unknown node, so obtaining an algebraic system which, once solved, will provide us the sought solution.

In order to develop a finite difference scheme, we start from Taylor expansion

$$u_{j+1} = u_j + \frac{\partial u}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 u}{\partial x^2} \Delta x^2 + \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^3 + O(\Delta x^4).$$

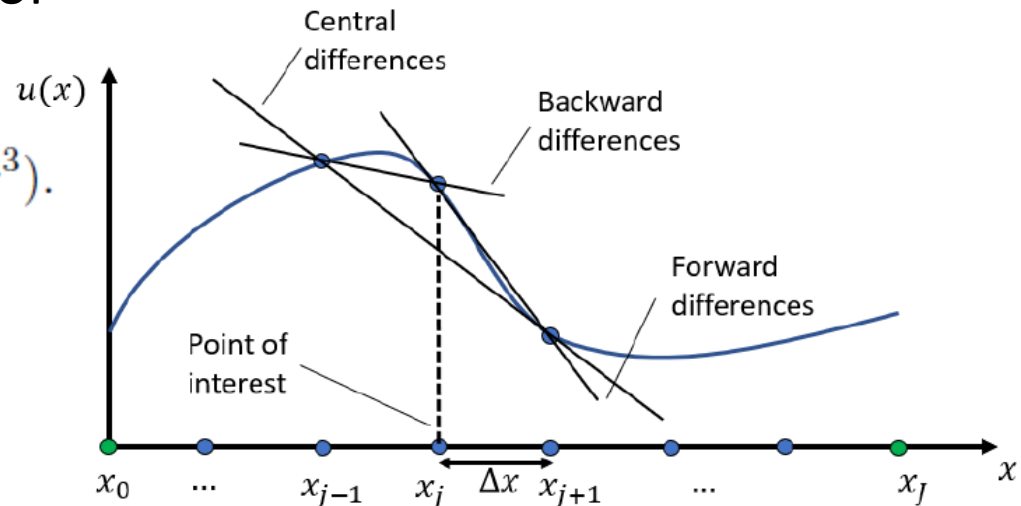
which, rearranged, leads to

$$\frac{u_{j+1} - u_j}{\Delta x} = \frac{\partial u}{\partial x} + \frac{1}{2} \frac{\partial^2 u}{\partial x^2} \Delta x + \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2 + O(\Delta x^3) = \frac{\partial u}{\partial x} + T.E.$$

where T.E. is the Truncation Error

$$T.E. = \frac{1}{2} \frac{\partial^2 u}{\partial x^2} \Delta x + \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2 + O(\Delta x^3).$$

This is the forward approximation of the first derivative.



We can repeat the same line of reasoning considering

$$u_{j-1} = u_j - \frac{\partial u}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 u}{\partial x^2} \Delta x^2 - \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^3 + O(\Delta x^4)$$

which rearranged

$$\frac{u_j - u_{j-1}}{\Delta x} = \frac{\partial u}{\partial x} - \frac{1}{2} \frac{\partial^2 u}{\partial x^2} \Delta x + \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2 + O(\Delta x^3).$$

leading to the **backward approximation** of the first derivative.

The leading term of the T.E. can be eliminated considering the sum of backward and forward approximations

$$\frac{1}{2} \left[\frac{u_j - u_{j-1}}{\Delta x} + \frac{u_{j+1} - u_j}{\Delta x} \right] = \frac{\partial u}{\partial x} + \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2 + O(\Delta x^4),$$

$$\frac{u_{j+1} - u_{j-1}}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2 + O(\Delta x^4).$$

which is the centred approximation of the first derivative.

To develop an **approximation formula for second order derivatives**, we subtract equations

$$\frac{u_{j+1} - u_j}{\Delta x} = \frac{\partial u}{\partial x} + \frac{1}{2} \frac{\partial^2 u}{\partial x^2} \Delta x + \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2 + O(\Delta x^3)$$



$$\frac{u_j - u_{j-1}}{\Delta x} = \frac{\partial u}{\partial x} - \frac{1}{2} \frac{\partial^2 u}{\partial x^2} \Delta x + \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2 + O(\Delta x^3).$$

leading to

$$\frac{u_{j+1} - u_j}{\Delta x} - \frac{u_j - u_{j-1}}{\Delta x} = \frac{\partial^2 u}{\partial x^2} \Delta x + O(\Delta x^3),$$

$$\frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2} = \frac{\partial^2 u}{\partial x^2} + O(\Delta x^2).$$

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}.$$

Forward Euler I (centred in space) CWE

We start by considering the pure linear advection equation

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} = 0.$$

At each node of the computational grid where the solution is unknown, we write using, for instance, central differences

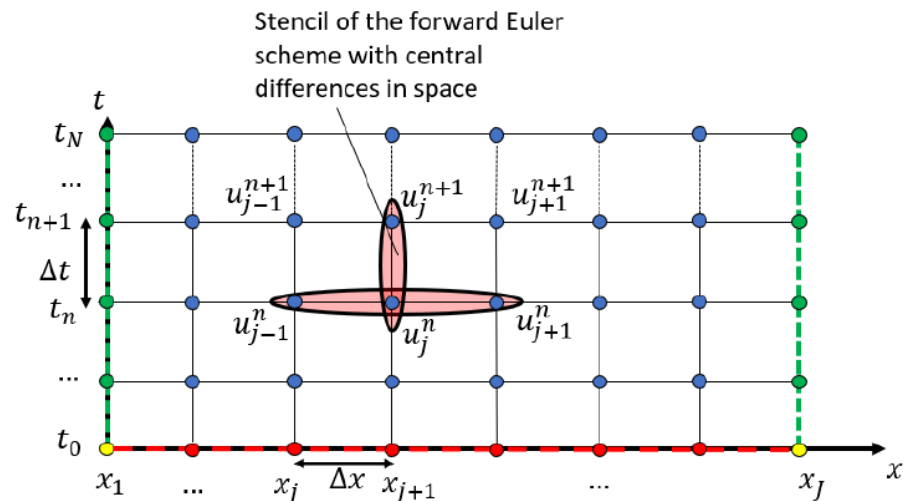
$$\left(\frac{\partial u}{\partial t} \right)_i^n = -\frac{U}{2\Delta x} (u_{i+1}^n - u_{i-1}^n),$$

where the apex n denotes the time at which quantities are considered. Assuming to calculate spatial derivatives at time n

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = -\frac{U}{2\Delta x} (u_{j+1}^n - u_{j-1}^n).$$

allowing to calculate

$$u_j^{n+1} = u_j^n - \frac{1}{2} \frac{U \Delta t}{\Delta x} (u_{j+1}^n - u_{j-1}^n),$$



Forward Euler II (centred in space) CWE

Writing such an equation for all unknown nodes, we obtain

$$\begin{bmatrix} u_1^{n+1} - u_1^n \\ u_2^{n+1} - u_2^n \\ u_3^{n+1} - u_3^n \\ \dots \\ u_{J-1}^{n+1} - u_{J-1}^n \\ u_J^{n+1} - u_J^n \end{bmatrix} = \begin{bmatrix} 0 & -Co/2 & 0 & \dots & 0 & 0 \\ Co/2 & 0 & -Co/2 & \dots & 0 & 0 \\ 0 & Co/2 & 0 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & -Co/2 \\ 0 & 0 & 0 & \dots & Co/2 & 0 \end{bmatrix} \begin{bmatrix} u_1^n \\ u_2^n \\ u_3^n \\ \dots \\ u_{J-1}^n \\ u_J^n \end{bmatrix}$$

being $Co = U\Delta t/\Delta x$ (**Courant Number**). The system can be written as

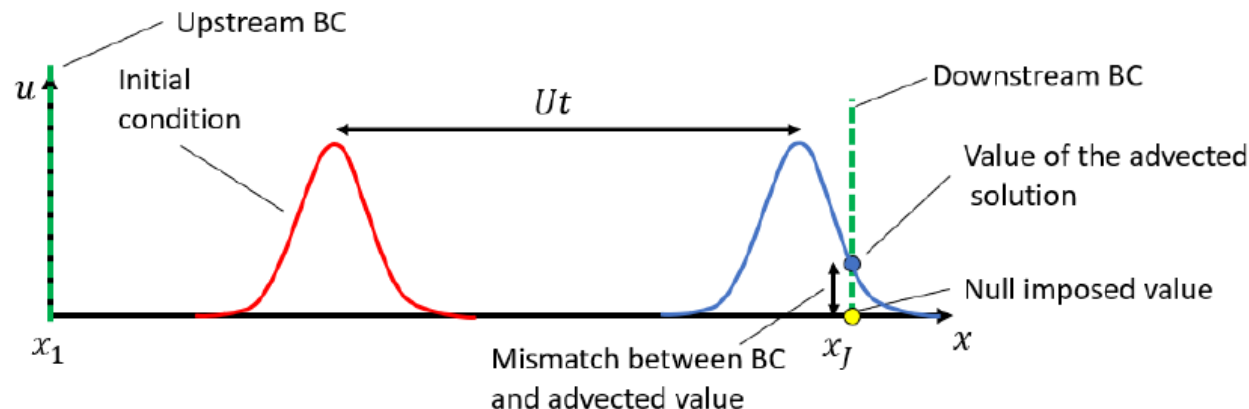
$$\begin{aligned} \mathbf{u}^{n+1} - \mathbf{u}^n &= Co \mathcal{A}_c \mathbf{u}^n. \\ \mathbf{u}^{n+1} &= [\mathbf{I} + Co \mathcal{A}_c] \mathbf{u}^n. \\ \mathbf{u}^{n+1} &= \mathcal{M}_{fc} \mathbf{u}^n. \end{aligned} \quad \mathcal{A}_c = \begin{bmatrix} 0 & -1/2 & 0 & \dots & 0 & 0 \\ 1/2 & 0 & -1/2 & \dots & 0 & 0 \\ 0 & 1/2 & 0 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & -1/2 \\ 0 & 0 & 0 & \dots & 1/2 & 0 \end{bmatrix}$$

where $\mathcal{M}_{fc}$ is an iteration matrix which transforms the old solution in the new one. **Unfortunately, the scheme is unstable.**

Actually, some of the values are known from the **boundary conditions**, so that

$$\begin{bmatrix} u_1^{n+1} \\ u_2^{n+1} \\ u_3^{n+1} \\ \dots \\ u_{J-1}^{n+1} \\ u_J^{n+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ Co/2 & 1 & -Co/2 & \dots & 0 & 0 \\ 0 & Co/2 & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & -Co/2 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{bmatrix} \begin{bmatrix} u(x_1) \\ u_2^n \\ u_3^n \\ \dots \\ u_{J-1}^n \\ u(x_J) \end{bmatrix},$$

It shall be noticed that fixing the value on the left-hand side is actually incompatible with the problem, as the value is set by the upstream solution.



Backward Euler (centred in space) CWE

In a similar way, we might calculate space derivatives **implicitly** at the new time-step

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = -\frac{U}{2\Delta x}(u_{j+1}^{n+1} - u_{j-1}^{n+1}).$$

which writing the full system leads to

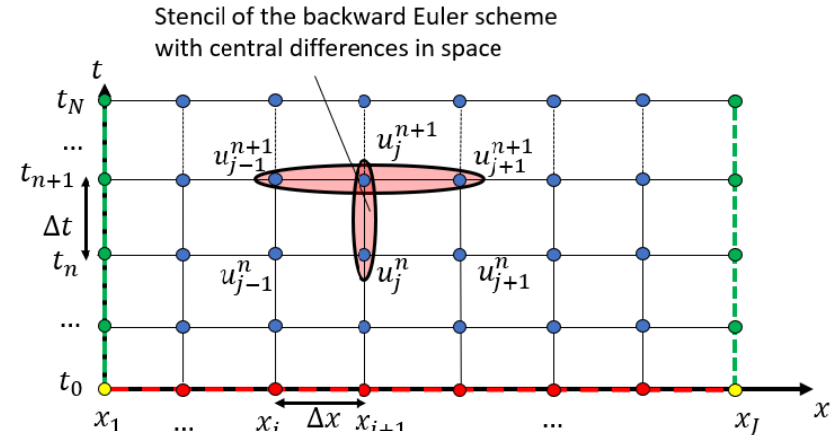
$$\begin{bmatrix} u_1^{n+1} - u_1^n \\ u_2^{n+1} - u_2^n \\ u_3^{n+1} - u_3^n \\ \vdots \\ u_{J-1}^{n+1} - u_{J-1}^n \\ u_J^{n+1} - u_J^n \end{bmatrix} = \begin{bmatrix} 0 & -Co/2 & 0 & \dots & 0 & 0 \\ Co/2 & 0 & -Co/2 & \dots & 0 & 0 \\ 0 & Co/2 & 0 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & -Co/2 \\ 0 & 0 & 0 & \dots & Co/2 & 0 \end{bmatrix} \begin{bmatrix} u_1^{n+1} \\ u_2^{n+1} \\ u_3^{n+1} \\ \vdots \\ u_{J-1}^{n+1} \\ u_J^{n+1} \end{bmatrix},$$

$$\mathbf{u}^{n+1} - \mathbf{u}^n = Co \, \mathcal{A}_c \mathbf{u}^{n+1}, \quad \longrightarrow \quad [\mathbf{I} - Co \, \mathcal{A}_c] \mathbf{u}^{n+1} = \mathbf{u}^n,$$

and finally

$$\mathbf{u}^{n+1} = [\mathbf{I} - Co \, \mathcal{A}_c]^{-1} \mathbf{u}^n = \mathcal{M}_{bc} \mathbf{u}^n,$$

Unfortunately, the scheme is very diffusive.



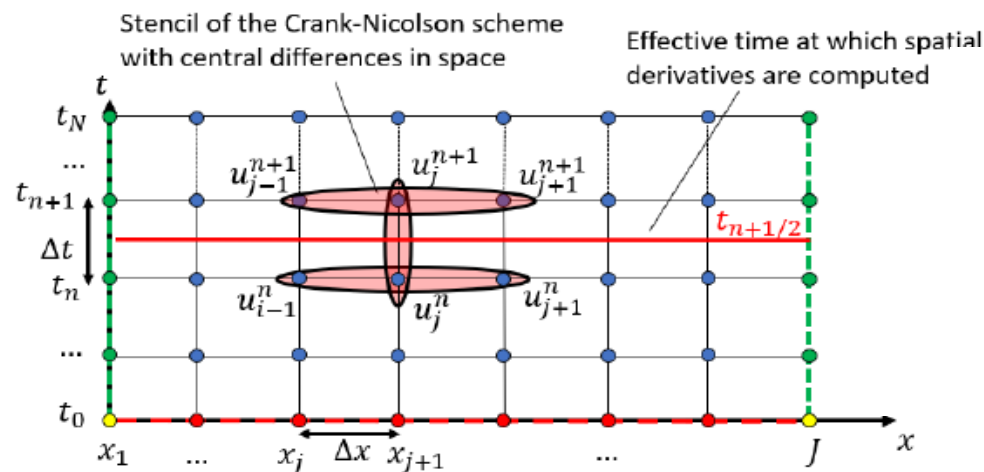
Crank-Nicolson (centred in space) CWE

It is also possible to calculate space derivatives (remember that they actually represent advective fluxes) at **an intermediate time**

$$\mathbf{u}^{n+1} - \mathbf{u}^n = Co \, \underline{\mathcal{A}}_c \left[\frac{\mathbf{u}^n + \mathbf{u}^{n+1}}{2} \right],$$

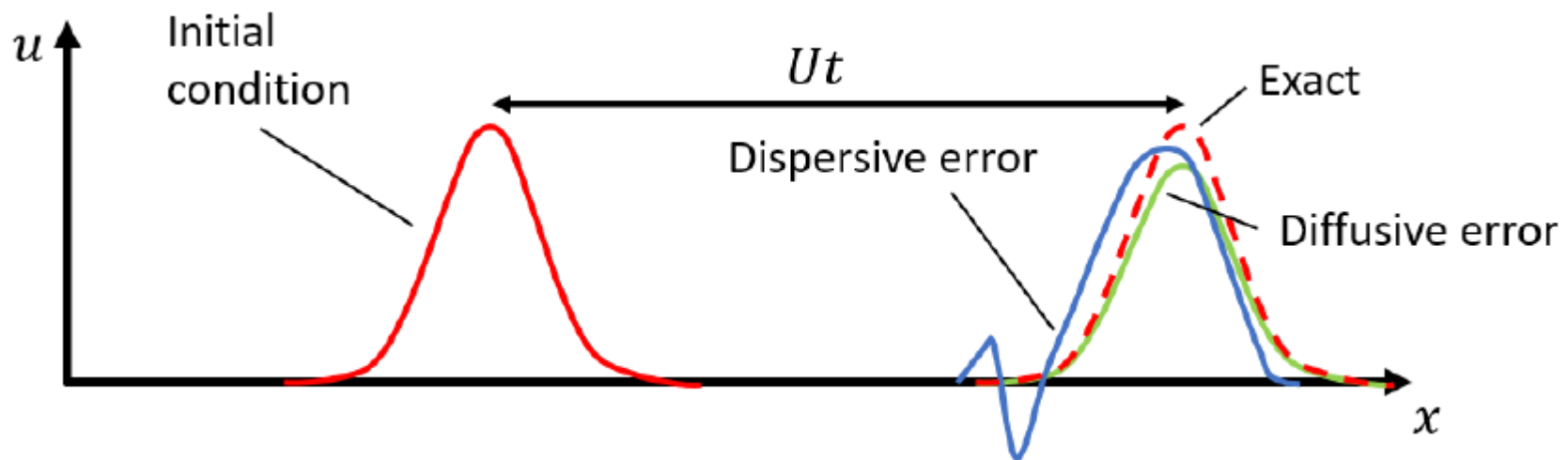
so that

$$\left[\underline{\mathcal{I}} - \frac{Co}{2} \underline{\mathcal{A}}_c \right] \mathbf{u}^{n+1} = \left[\underline{\mathcal{I}} + \frac{Co}{2} \underline{\mathcal{A}}_c \right] \mathbf{u}^n, \quad \Rightarrow \quad \mathbf{u}^{n+1} = \left[\underline{\mathcal{I}} - \frac{Co}{2} \underline{\mathcal{A}}_c \right]^{-1} \left[\underline{\mathcal{I}} + \frac{Co}{2} \underline{\mathcal{A}}_c \right] \mathbf{u}^n,$$



The scheme is characterized by stability and low diffusion, it is however dispersive.

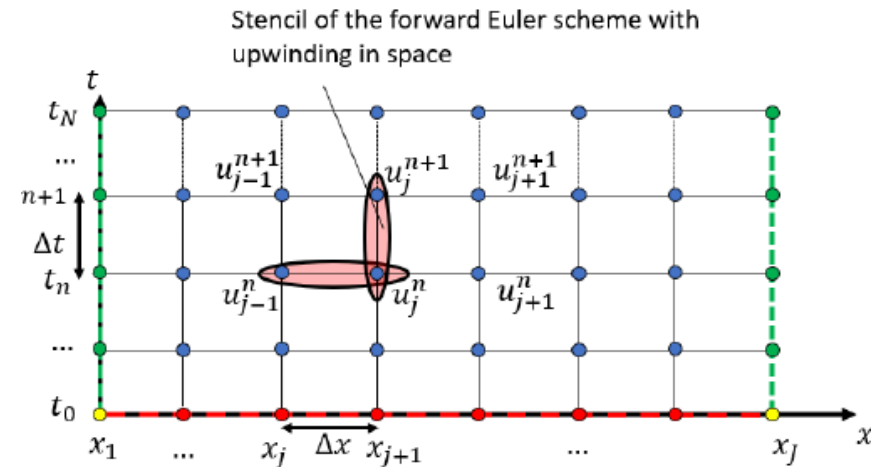
We mentioned that the Crank-Nicolson scheme with central differences is not diffusive but it is dispersive. Without entering into details, it is good to have a qualitative understanding of the difference between the two



Navier-Stokes equations have both a diffusive and dispersive behaviour, so that **the error introduced by the numerical scheme becomes often undistinguishable from the real solution.**

It is also possible to calculate space derivatives with an upwind stencil

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = -\frac{U}{\Delta x}(u_j^n - u_{j-1}^n).$$



$$\begin{bmatrix} u_1^{n+1} - u_1^n \\ u_2^{n+1} - u_2^n \\ u_3^{n+1} - u_3^n \\ \dots \\ u_{J-1}^{n+1} - u_{J-1}^n \\ u_J^{n+1} - u_J^n \end{bmatrix} = \begin{bmatrix} -Co & 0 & 0 & \dots & 0 & 0 \\ Co & -Co & 0 & \dots & 0 & 0 \\ 0 & Co & -Co & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & -Co & 0 \\ 0 & 0 & 0 & \dots & Co & -Co \end{bmatrix} \begin{bmatrix} u(x_1) \\ u_2^n \\ u_3^n \\ \dots \\ u_{J-1}^n \\ u(x_J) \end{bmatrix} \quad \mathcal{A}_u = \begin{bmatrix} -1 & 0 & 0 & \dots & 0 & 0 \\ 1 & -1 & 0 & \dots & 0 & 0 \\ 0 & 1 & -1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & -1 & 0 \\ 0 & 0 & 0 & \dots & 1 & -1 \end{bmatrix}$$

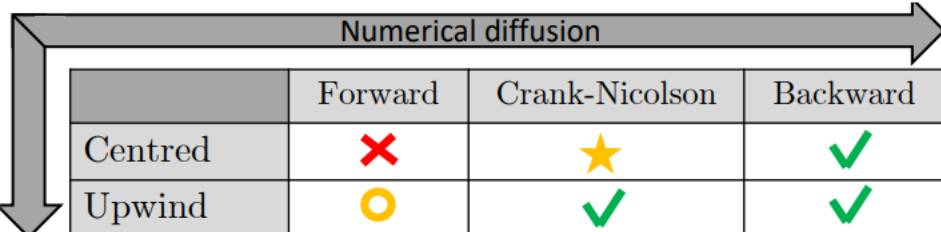
$$\mathbf{u}^{n+1} - \mathbf{u}^n = Co \mathcal{A}_u \mathbf{u}^n. \quad \longrightarrow \quad \mathbf{u}^{n+1} = [\mathbf{I} + Co \mathcal{A}_u] \mathbf{u}^n.$$

The scheme is **conditionally stable**, i.e. its stability depends on Courant, which must be so that $Co < 1$.

Summarizing, by changing how derivatives are approximated, we obtain different behaviours of the numerical schemes, which manifest in

- different accuracy;
- different stability conditions.

For the pure linear advection equation, the stability for various numerical schemes is reported below



	Numerical diffusion		
	Forward	Crank-Nicolson	Backward
Centred	✗	★	✓
Upwind	○	✓	✓

✓ Unconditionally stable

★ Neutrally stable

○ Conditionally stable

✗ Unconditionally unstable

It shall be noticed that stability conditions depend on the considered equation, and that both space and time schemes contribute to define stability and accuracy.

We now need to better understand the effect induced by the choice of the numerical scheme. In particular

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} = 0.$$

Adopting the following approximations

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} \approx \frac{\partial u}{\partial t} + \frac{1}{2} \frac{\partial^2 u}{\partial t^2} \Delta t.$$

$$\frac{u_{j+1} - u_{j-1}}{2\Delta x} \approx \frac{\partial u}{\partial x} + \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2.$$

The actual equations we are solving for, considering only the leading term of the T.E. is thus

$$\frac{\partial u}{\partial t} + \frac{1}{2} \frac{\partial^2 u}{\partial t^2} \Delta t + U \left[\frac{\partial u}{\partial x} + \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2 \right] = 0.$$

Taking T.E related terms on the right

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} = - \left[\frac{1}{2} \frac{\partial^2 u}{\partial t^2} \Delta t + U \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2 \right],$$

However, the original equation has been modified only by small terms, so that approximately

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} = 0. \quad \longrightarrow \quad \frac{\partial u}{\partial t} = -U \frac{\partial u}{\partial x}.$$

so that

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} = - \left[\frac{1}{2} \frac{\partial^2 u}{\partial t^2} \Delta t + U \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2 \right],$$

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} = \frac{1}{2} \frac{\partial}{\partial t} \left[U \frac{\partial u}{\partial x} \right] \Delta t - U \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2,$$

Leading to

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} = \frac{U}{2} \frac{\partial^2 u}{\partial x \partial t} \Delta t - U \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2.$$

Considering again the original differential equation

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} = 0. \quad \longrightarrow \quad \frac{\partial^2 u}{\partial x \partial t} = -U \frac{\partial^2 u}{\partial x^2}$$

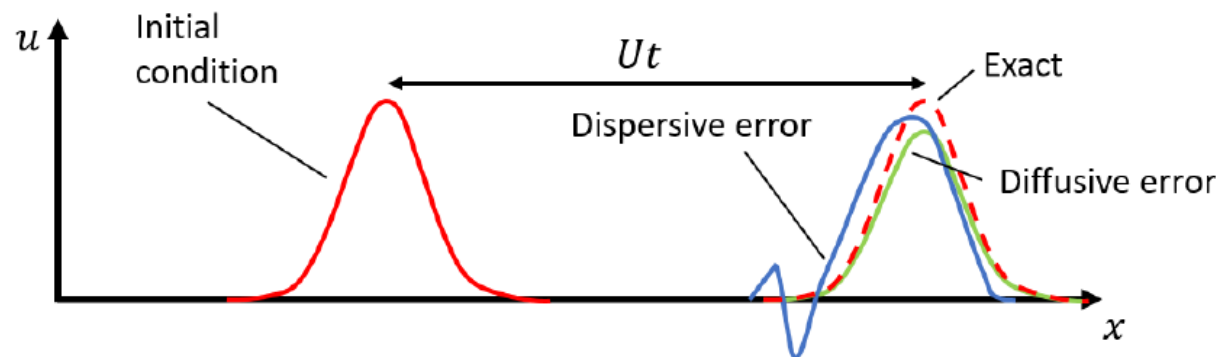
Finally, leading to

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} = \boxed{\text{Diffusion error}} - \frac{U^2}{2} \frac{\partial^2 u}{\partial x^2} \Delta t - \boxed{\text{Dispersive error}} + U \frac{1}{6} \frac{\partial^3 u}{\partial x^3} \Delta x^2.$$

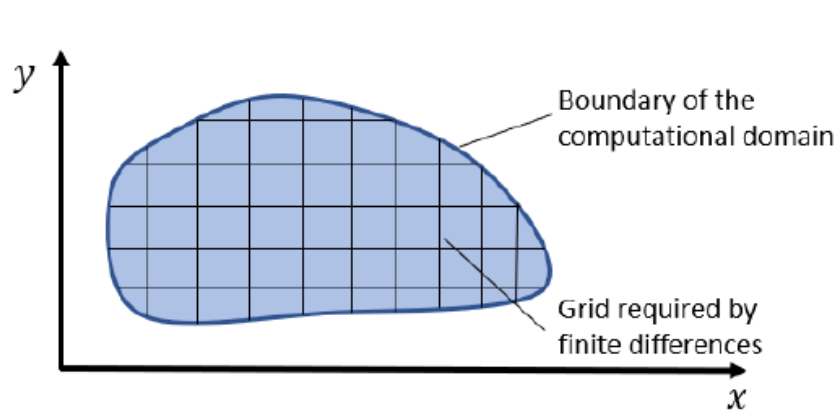
which is the **actual differential equation** solved by the discretized **system** (considering only the leading T.E.).

As it can be seen, starting from linear advection, we arrived to a linear advection-diffusion equation. **In this case diffusion is negative, causing the scheme to be unstable.** We thus introduce the concept of numerical viscosity

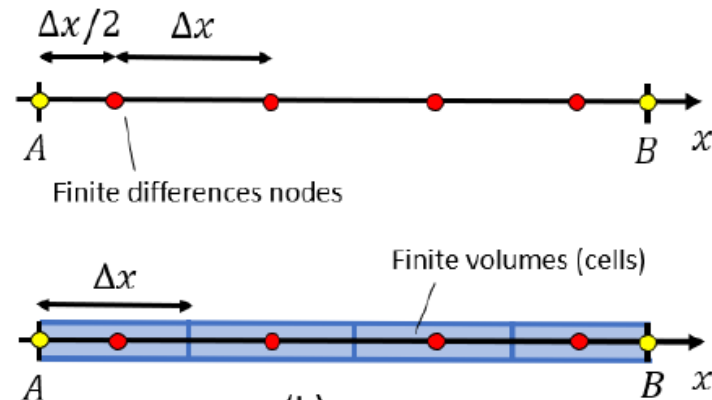
$$\nu_{num} = -\frac{U^2}{2} \Delta t,$$



Finite differences are not well adapted to problems characterized by complex geometrical boundaries

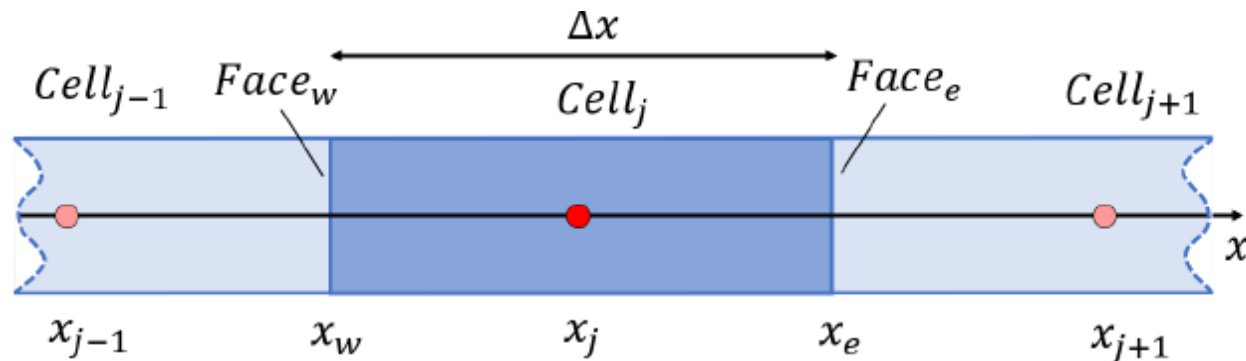


(a)



(b)

To develop the Finite Volume method, we firstly need to subdivide the domain in small chunks, denoted as **cells**.



We integrate the governing equation over the cell

$$\int_{cell_j} \left[\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} - \nu \frac{\partial^2 u}{\partial x^2} \right] dx = 0.$$

The first term can be approximated as

$$\int_{cell_j} \frac{\partial u}{\partial t} dx = \frac{d}{dt} \left[\int_{cell_j} u dx \right] \approx \frac{d(u_j \Delta x)}{dt} = \frac{du_j}{dt} \Delta x,$$

The second one is

$$\int_{cell_j} U \frac{du}{dx} dx = U[u(x_e) - u(x_w)].$$

In which we might approximate

$$u(x_e) = \frac{u_{j+1} + u_j}{2}, \quad u(x_w) = \frac{u_{j-1} + u_j}{2},$$

As regard the last term

$$\int_{cell_j} \nu \frac{\partial^2 u}{\partial x^2} dx = \nu \left[\frac{\partial u}{\partial x} \Big|_{x_e} - \frac{\partial u}{\partial x} \Big|_{x_w} \right].$$

which can be approximated as

$$\frac{\partial u}{\partial x} \Big|_{x_e} \approx \frac{u_{j+1} - u_j}{\Delta x}, \quad \frac{\partial u}{\partial x} \Big|_{x_w} \approx \frac{u_j - u_{j-1}}{\Delta x}.$$

Adding everything together

$$\frac{du_j}{dt} \Delta x + U \left[\frac{u_{j+1} + u_j}{2} - \frac{u_{j-1} + u_j}{2} \right] - \nu \left[\frac{u_{j+1} - u_j}{\Delta x} - \frac{u_j - u_{j-1}}{\Delta x} \right] = 0.$$

which is **equivalent to a centred finite difference formulation**

$$\frac{du_j}{dt} + U \left[\frac{u_{j+1} - u_{j-1}}{2\Delta x} \right] - \nu \left[\frac{u_{j-1} - 2u_j + u_{j+1}}{\Delta x^2} \right] = 0.$$

We might also calculate fluxes using the upwind cell values

$$\frac{du_j}{dt} + U \left(\frac{u_{j+1} - u_{j-1}}{2\Delta x} \right) - \nu \left(\frac{u_{j-1} - 2u_j + u_{j+1}}{\Delta x^2} \right) = 0.$$

Leading to a formulation identical to a finite difference scheme based on an upwind approximation of first order derivatives.

The Finite Volume method is the work horse of Computational Fluid Dynamics.